

Taylor McDonald

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Data Science student at NJIT building ML-powered full-stack applications in predictive modeling and real-time data systems.
Seeking a Summer 2026 internship in data science or ML engineering.

SKILLS

Programming Languages: Python, SQL

ML/DS: scikit-learn, XGBoost, pandas, numpy, PyTorch (basics)

Tools: Git, FastAPI, Jupyter, VS Code, Hugging Face Spaces

Areas: Machine Learning, Statistical Learning/Modeling, Regression Analysis, Data Wrangling, Model Deployment

Spoken Languages: English(fluent), Korean (intermediate)

PROJECTS

PawsIQ — Pet Services Intelligence Platform (In Progress) | [Github](#)

April 2026 – Present

- Developing demand forecasting model using GradientBoosting on 9,935 synthetic NJ bookings, achieving 5% RMSE improvement over mean baseline with stable cross-validation MAE of 0.774 ± 0.033 across 5 time-series folds
- Engineering dynamic pricing pipeline using Ridge Regression ($R^2=0.727$) to predict surge multipliers per booking, producing 16.7% simulated revenue lift over flat-rate baseline across 1,856 held-out test bookings
- Designing FastAPI backend serving both ML models as live REST endpoints; built React operator dashboard visualizing demand heatmap, surge pricing schedule, and walker performance across 8 walkers and 9,935 bookings
- Stack: Python, scikit-learn, pandas, FastAPI, React, Vercel

House Prices Prediction | [Github](#)

March 2026

- Engineered 5 features (TotalSF, TotalBathrooms, HouseAge, OverallScore, binary flags) from 79 raw variables using Random Forest importance, reducing noise and improving model signal
- Trained and compared 3 models (Linear Regression, KNN, Random Forest) on 1,460 samples, achieving R -squared=0.87 and RMSE=0.14 with Random Forest - a 22% RMSE improvement over the Linear Regression baseline
- Cleaned and imputed missing values across 79 features, producing a fully reproducible pipeline from raw Kaggle CSV to final predictions
- Stack: Python, pandas, numpy, scikit-learn, matplotlib, seaborn

FocusFit (In Progress) | [Github](#)

Dec 2025 - Present

- Developing ML model to predict optimal study session lengths from user focus pattern data
- Engineered features from session logs using pandas; evaluated classifier performance via accuracy and F1 score
- Visualized productivity trends with matplotlib to surface actionable insights
- Stack: Python, pandas, scikit-learn, matplotlib

WORK EXPERIENCE

All The Paws | Founder | East Orange, NJ

Sep 2025 - Present

- Founded and registered a pet services business, growing to 15+ recurring clients within months by building trust through consistent, high-quality service delivery
- Designed and implemented end-to-end booking and scheduling systems - covering invoicing, daily care routines, and emergency protocols - reducing operational friction across all client accounts
- Built and maintain the business website, managing branding, service copy, and client-facing UX to support client acquisition

EDUCATION

Data Science B.S. | New Jersey Institute of Technology, Newark, NJ

Expected May 2027

GPA: 3.1 | Sep 2024 - Present

Coursework: Intro to ML, Regression Analysis, Statistical Methods I and II, Statistical Methods in Data Science, Multivariate Distribution, Linear Algebra

Study Abroad | Hanyang University ERICA, Ansan, South Korea

Feb 2025 - Jun 2025

Coursework: Advanced Calculus, Intensive Korean

HONORS AND ACTIVITIES

JPMorgan Chase and Co. Career.edYOU Academy Graduate | 2023

Headstarter AI Software Engineering Fellow | 2024

Community Service Certificate | Mission Trip, South Korea | 2024

Onyx Dance Crew | NJIT | 2025

CERTIFICATIONS

Python Certification (In Progress) | freeCodeCamp

January 2026

Supervised Machine Learning: Regression and Classification | Coursera

February 2025

Foundations: Data, Data, Everywhere | Coursera

May 2024